

Order reduction, simplification and parameters identification for cold start model of PEM fuel cell

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ABSTRACT

The cold-start of proton exchange membrane fuel cell (PEMFC) has been one of the leading technical challenges for fuel cell vehicle commercialization. It is essential to understand the mechanism and process of a cold start to improve the cold start ability of the PEM fuel cell. In this study, a one-dimensional cold start transient model of PEMFC was developed for transfer species, heat, charge and water phase change during the cold start process. Based on the 1-D spatially distributed model, a reduced nonlinear cold-start model of the PEMFC is presented. Then the parameter identification is conducted to validate the model accuracy based on the experiment data. The identification result shows that the reduced model captures the detailed model's multiplicity behaviours, and good agreement is observed during cold start simulation with experiment results. In contrast, the computation time is reduced significantly. Thus, the proposed order reduction model can be used to investigate real-time estimation and develop model-based control strategies for automotive PEMFC cold start.

1. Introduction

Proton exchange membrane (PEM) fuel cells have the advantages of high power density, high efficiency, low operating temperature, and zero emissions, making them one of the most promising alternatives for the new generation of automotive powertrains. However, low-temperature cold start capability has become one of the main obstacles hindering the commercial application of fuel cell vehicles [1–3]. The U.S. Department of Energy (DOE) has proposed PEM fuel cell cold start goals by 2020, with a self-cold start at -30°C , assisted cold start at -40°C , and 50 % of rated power within 30 s at -20°C [4]. Over the past decade, PEM fuel cell cold start technology has made significant advances. Fuel cell vehicles, including the Toyota Mirai and Honda Clarity, have already met the Department of Energy's cold targets [5–7]. However, the relationship between cold-start performance and internal water-thermal processes remains unclear.

In the cathode catalyst layer (CCL) of PEM fuel cells, hydrogen and oxygen react to produce water, simultaneously releasing electricity and heat. The water is crucial for the polymer electrolyte membrane, which

is hydrated by the membrane and improves proton conductivity. However, the water produced in the CCL changes into ice at subzero temperatures. The ice occupied the porous medium, which hindered the oxygen transport and reduced the active surface of the catalyst. Therefore, the temperature rise and the ice formation rate are the main issues during the cold start process. The PEM fuel cell must be heated to zero degrees before the porous medium, and electrochemical reaction area are blocked by ice to achieve a successful cold start.

In previous research on the cold start of the fuel cell, many experiments have been conducted to investigate cold start performance and internal process. Tajiri et al. [8,9] conducted the isothermal cold start experiments on a 25 cm^2 single fuel cell. Initial temperature and membrane water content were the main factors affecting cold start performance. The higher the initial temperature and lower membrane water content, the longer the survival time of the PEM fuel cell. Thompson et al. [10,11] found that the catalyst layer's ice formation will intensify current density distribution inconsistently and reduce the water storage capacity for the high load current. High-frequency resistance (HFR) is essential for polymer membrane hydration during the cold start.

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Table 1
Model parameters.

Model parameter	Value
Thickness of B.P.; channel; GDL; CL; membrane	$\delta_{BP,CH,GDL,CL,MEM} = 1050, 300, 215, 10, 18\mu\text{m}$
Thickness of polar plate; current collector; insulation plate	$\delta_{pp,cc,isp} = 4.15, 3.2, 9\text{mm}$
Active area	$A_{fc} = 25,330\text{cm}^2$
Single cell anode and cathode channel volume	$V_{an} = 1.08\text{E} - 6\text{m}^3; V_{ca} = 1.08\text{E} - 6\text{m}^3$
Stack anode and cathode channel volume	$V_{an} = 4.128\text{E} - 6\text{m}^3; V_{ca} = 6.191\text{E} - 6\text{m}^3$
Single cell anode and cathode orifice coefficient	$k_{ca_out} = 5\text{E} - 11\text{kg/s/Pa}; k_{an_out} = 2\text{E} - 11\text{kg/s/Pa};$
Stack anode and cathode orifice coefficient	$k_{ca_out} = 5\text{E} - 9\text{kg/s/Pa}; k_{an_out} = 2\text{E} - 9\text{kg/s/Pa};$
Molar mass of oxygen, hydrogen water	$M_{O_2,H_2,H_2O} = 32, 2, 18\text{ g/mol}$
Liquid water and ice mass density	$\rho_{lq,ice} = 990, 920\text{kg/m}^3$
Density of CL, GDL, B.P.	$\rho_{CL,GDL,BP} = 1230, 1230, 2600\text{kg/m}^3$
Density of polar plate, current collector and insulation plate	$\rho_{pp,cc,isp} = 2811, 8960, 2200\text{kg/m}^3$
Membrane density; equivalent weight	$\rho_m = 2010\text{kg/m}^3; EW = 976\text{g/mol}$
Volume fraction of ionomer in CL; the porosities of CL; GDL;	$\omega = 0.15; \epsilon_{cl,gdl} = 0.4, 0.6$
Contact angles of CL and GDL	$\theta_{CL,GDL} = 100^\circ, 120^\circ$
Intrinsic permeability of CL; GDL	$K_{CL,GDL} = 6.2 \times 10^{-13}, 6.2 \times 10^{-12}\text{m}^2$
Specific heat capacity of membrane, CL, GDL, B.P.	$(c_p)_{m,gdl,cl,bp} = 833, 750, 750, 808\text{J/kg/K}$
Specific heat capacity of polar plate, current collector and insulation plate	$(c_p)_{pp,cc,isp} = 808, 394, 750\text{J/kg/K}$
Thermal conductive of membrane, CL, GDL, B.P.	$k_{mem,cl,gdl,bp} = 0.95, 1.2, 1.5, 20\text{W/m/K}$
Thermal conductive of polar plate, current collector and insulation plate	$k_{pp,cc,isp} = 20, 386.4, 0.25\text{W/m/K}$
Latent heat of water	$h_{cond} = 235\text{kJ/kg}; h_{fussn} = 335\text{kJ/kg}$

Table 1 (continued)

Model parameter	Value
condensation and fusion	
Hydrogen diffusivity coefficient	$D_{H_2} = 1.005 \times 10^{-4} (T/333.15)^{1.5} \text{m}^2/\text{s}$
Oxygen diffusivity coefficient	$D_{O_2} = 2.652 \times 10^{-5} (T/333.15)^{1.5} \text{m}^2/\text{s}$
Oxygen and hydrogen reference concentration	$c_{O_2,ref} = 40\text{mol/m}^3; c_{H_2,ref} = 40\text{mol/m}^3$
Charge transport coefficients	$\alpha_a = \alpha_c = 1$
Pt load in anode and cathode	$I_{pt}^{an} = 0.1\text{mg/cm}^2; I_{pt}^{ca} = 0.35\text{mg/cm}^2$
Reference exchange current density for HOR and ORR	$i_{ref}^{ca} = 2.8\text{E} - 4\text{ A/m}^2; i_{ref}^{an} = 300\text{A/m}^2$
Electrochemical active surface area	$ECSA^{an} = 15[\text{m}^2/\text{gPt}]; ECSA^{ca} = 50[\text{m}^2/\text{gPt}]$
Electric conductivities	$\sigma_{CL,GDL,BP} = 300, 300, 20000\text{S/m}$

According to Tabe and Charles et al. [12–14] PEM fuel cell cold start experiments, the HFR decreased rapidly at the initial stage and increased until the cold start failed. It indicated that the ionomer hydrated water at the initial stage, which improved the conductivity of the polymer membrane. After the membrane water reached saturation, the ice formation at the interface between CL and GDL increased the contact resistance. The electrochemical impedance spectroscopy (EIS) is an effective technical tool for investigating the electrochemical reaction and species transport process [15–18]. Meanwhile, cyclic voltammetry (CV) is always used to investigate the electrochemical active surface area (ECSA). Mukundan and Wang et al. [19–21] reported that the ECSA remained constant during the cold start, while Ge et al. [22] pointed out that a decreasing trend exists. The Contradiction phenomenon has been observed during past research. Wang and Oszcipok et al. [19,23,24] conducted the EIS test at a small load current to reach a long survival time during a cold start. The experiment's result shows that the ohmic and charge transfer resistances remain almost constant. In contrast, the mass transfer resistance increases dramatically due to the ice blocking the reaction gas transport. Supercooled water is observed during the PEM fuel cell cold start experiment [25–28]. The supercooled water is metastable, which will be changed immediately by agitation and vibration. Biesdorf et al. [26,28] pointed out that the supercooled water stochasticity would decrease considerably when the fuel cell area is larger and the cold start temperature is lower. So large surface PEM fuel cell cold start below -10°C can safely neglect the stochasticity of supercooled water.

Model is an effective tool to investigate the internal process of PEM fuel cell subzero start-up. Mao [29] first established the 3-D numerical cold-start model to explore the current density distribution, membrane water content, temperature and ice in the catalyst layer. Jiao et al. [30,31] studied various operating and initial conditions on cold start performance based on nonequilibrium phase change. Nevertheless, the phase change coefficient is constant during the cold start process. Huo et al. [32] studied the effect of different phase change coefficients on cold start failure time for isothermal cold start. Newman et al. [33–35] built the 2-D numerical model while the water phase change rates are developed based on the chemical potential difference between liquid water, vapour and ice.

Three-dimensional numerical model can describe the distribution of internal states and the electrochemical reaction process. Therefore, they

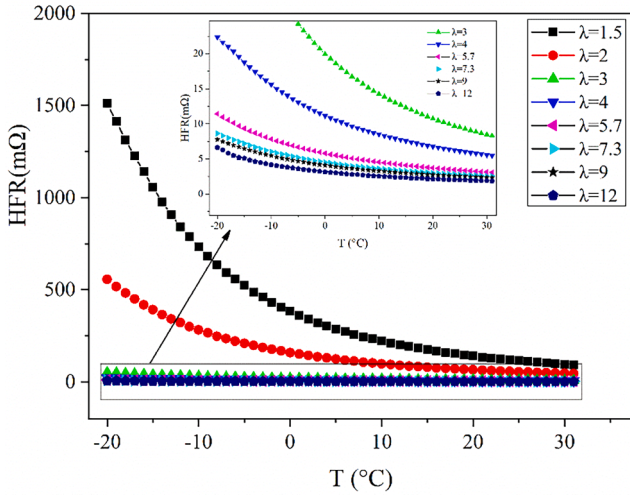


Fig. 1. HFR versus temperature for membrane at various water content.

required high computational effort even for a small sandwich PEM fuel cell domain. Macedo-Valencia et al. [36] built a 3-D CFD model for a six-cell stack with an area of 31.2 cm², which took almost 24 h to solve the model in the workstation. So, solving a vehicle PEM fuel cell stack model is almost impossible. In order to develop a control system for a rapid cold start strategy, a lower-dimensional model is required to realize real-time control. Yun et al. [37,38] developed the analytical model to investigate the cold start process. A successful cold start was divided into three stages: membrane hydration, ice formation, temperature rise, and ice melting. In addition, the impact of ice fraction and tortuosity of the catalyst layer on cell performance was studied. However, the membrane water transport and phase change were ignored here. Fang et al. [39] developed the model order reduction method, and then the reduction model is used to develop a real-time control system. However, the simplification model focuses on room temperature, while the mathematical model is unsuitable for PEM fuel cell sub-frozen cold start research. Mcai and Mangold et al. [40–42] developed a control-oriented PEM fuel cell dynamical model for room temperature two-phase flow. The control volume method was used to discretize the model domain. However, the catalyst layer was ignored while conceded as the interface between the membrane and GDL domain, which would bring errors in describing the catalyst layer species transport.

In past studies, the order reduction and simplification of the PEM fuel cell cold-start model were hard to find in the literature. Jiang et al. [43] established the cold start model and simplified the numerical model based on the control volume method. However, the phase change rate during the cold start process was constant, while the supercooled water in the CL and GDL was not considered here. Jiao et al. [44] built the 1 + 1D multiphase real-time PEM fuel cell model under various operating conditions. Although the supercooled water was considered under sub-freezing temperature, the phase change rate was set constant here, and the model was not validated enough for different cold start operation conditions. Here we derive a reduced-order cold-start model of a PEM fuel cell based on the control volume method for control design purposes. The phase transition between supercooled water and ice is considered here, and a phase transition model of supercooled water is established based on the classical nucleation theory. Then the simplification cold start model is validated with isothermal and non-isothermal cold start experiments under various operating conditions. The paper is organized as follows. A one-dimensional numerical model is established in Section 2, and then the model is reduced and simplified in Section 3. Parameter identification is made in Section 4. In Section 5, the calibration of the model and the verification of the experimental results are presented.

2. 1-D numerical model

In this section, a 1-D transient cold-start model of PEM fuel cells is established according to the previous studies [10,11,32,45]. The main physical and chemical processes considered in the model are given here, including electrochemical reactions and electric charges transport, mass transfer, phase transition, and heat transfer. Only the direction perpendicular to the MEA (through-plane) is considered here. The primary geometry and physic parameters for the present modelling work are given in Table 1.

2.1. Electrochemical reaction and charge transport

The hydrogen oxidation and oxygen reduction reactions occur in the anode and cathode CL. The electrode reaction kinetic is usually delineated by the Butler-Vomer equation (B-V equation), wherever the exchange current density at the anode and cathode CL is described as

$$j_a = (1 - s_{ice} - s_{li})^{\tau_a} j_{0,a} \left(\frac{c_{H_2}}{c_{H_2,ref}} \right)^{0.5} \left[\exp\left(\frac{\alpha_a F \eta}{RT}\right) - \exp\left(-\frac{\alpha_c F \eta}{RT}\right) \right] \quad (1)$$

$$j_c = (1 - s_{ice} - s_{li})^{\tau_c} j_{0,c} \frac{c_{O_2}}{c_{O_2,ref}} \left[\exp\left(\frac{\alpha_a F \eta}{RT}\right) - \exp\left(-\frac{\alpha_c F \eta}{RT}\right) \right] \quad (2)$$

Here s_{ice} and s_{li} is the ice and liquid water saturation in the CL, τ_a is the ECSA corrected parameter affected by the ice and liquid water saturation. $j_{0,a}$ and $j_{0,c}$ are the exchange current density of the anode and cathode catalyst, which are described as $j_{0,i} = i_{ref} * ECSA * Lpt$. Here i_{ref} is the reference exchange current density for anode and cathode, ECSA is the electrochemical active surface area of catalyst. Lpt is the Pt load in anode and cathode. c_{H_2} and c_{O_2} are the local hydrogen and oxygen concentration in CL. While $c_{H_2,ref}$ and $c_{O_2,ref}$ are the hydrogen and oxygen reference concentrations. Furthermore, α_a and α_c are charge transport coefficients of electrochemical reaction. η is the overpotentials defined as

$$\eta = \phi_s - \phi_e - E_{eq} \quad (3)$$

Here ϕ_s is the solid electrode potential, ϕ_e refers to electrolyte phase potential. E_{eq} is the electrode reaction equilibrium potential, which is zero at the anode. However, the Nernst equation describes the equilibrium potential in the cathode as

$$E_{eq} = 1.23 - 0.9 \times 10^{-3} \cdot (T - T_0) + \frac{RT}{2F} \ln \left(\frac{p_{H_2} p_{O_2}^{0.5}}{p_{H_2O}} \right) \quad (4)$$

The charge conservation equation describes the protons and electrons transport in the PEM fuel cell as

$$-\nabla \cdot (\sigma_s^{eff} \nabla \phi_s) = j \quad (5)$$

$$-\nabla \cdot (\sigma_e^{eff} \nabla \phi_e) = -j \quad (6)$$

Here σ_s^{eff} and σ_e^{eff} are effective electronic and ionic conductivity, modified in step with the volume fraction of the ionomer and the porosity of the medium.

$$\sigma_s^{eff} = (1 - \varepsilon - \omega)^{1.5} \sigma_s \quad (7)$$

$$\sigma_e^{eff} = \omega^{1.5} \sigma_m \quad (8)$$

Here ω is the ionomer volume fraction in CL and ε is the CL and GDL porosities. The ionomer conductivity is heavily related to the water content and temperature of the membrane. Here we experiment to measure the ionomer ionic conductivity. The PEM fuel cell is revitalized at 100 % humidity constant current density for half an hour, and then an equilibrium purge is conducted to reach a constant water content in the ionomer. Then the PEM fuel cell is frozen to set temperature while the HFR (high-frequency resistance) is recorded. Different membrane contents' high-frequency resistance varies with temperatures as Fig. 1

Table 2
Source terms of gas species.

Gas species	ACL	CCL	GDL
H ₂	$S_{H_2} = -\frac{M_{H_2}j}{2F}$	0	0
O ₂	0	$S_{O_2} = -\frac{M_{O_2}j}{4F}$	0

shows.

We fitted the measured membrane HFR data with an exponent polynomial in λ to obtain a smooth parameterization conductivity that captures all essential features of the data:

$$\sigma_m = \exp(5.496 - 9.371/\lambda^{0.3978}) \cdot \exp\left(\frac{32270}{R} \left(\frac{1}{273} - \frac{1}{T}\right)\right) \quad (9)$$

2.2. Governing equations

The cold start process of the PEM fuel cell is conducted at a temperature range from -20°C to 0°C . The saturated vapour pressure is defined as:

$$\log_{10}\left(\frac{P_{\text{sat}}}{101325}\right) = -2.1794 + 0.02953(T - 273.15) - 9.1837 \times 10^5 (T - 273.15)^2 + 1.4454 \times 10^{-7} (T - 273.15)^3 \quad (10)$$

During the temperature range, the max saturated vapour pressure is 675 Pa. So, the vapour concentration among the fuel cell domain is much tiny. According to Thompson[11] etc. experiment studies, the predried membrane sample ($\lambda \sim 1$) was placed in an environmental chamber at -20°C . After several days of exposure, the membrane water content reached almost 3 or 4, suggesting that the water uptake rate from the saturated vapour phase at low temperature is relatively low. Dry gas was supplied to the fuel cell during the cold start process, and the membrane water uptake and desorb rate were relatively low. Therefore, we can safely neglect the vapour concentration transport among the fuel cell domain.

The mass conservation of hydrogen and oxygen gas transport in the fuel cell is described as:

$$\frac{\partial(\varepsilon(1 - s_{\text{liq}} - s_{\text{ice}})C_i)}{\partial t} = \frac{\partial^2(D_i^{\text{eff}}C_i)}{\partial x^2} + S_i \quad (11)$$

Here C_i is the hydrogen and oxygen molar concentration, while D_i^{eff} is the hydrogen and oxygen effective diffusivities coefficient. s_{liq} and s_{ice} are liquid water and ice saturation. The gas species source terms determined by the electrochemical reaction are shown in Table 2.

The bulk porosity and the volume fraction of ice and liquid water affect the gas effective diffusivity coefficient. The effective diffusivity coefficient for gas transport in CL is described by power-law exponents [46].

$$D_{\text{CL}}^{\text{eff}} = (\varepsilon_0(1 - s_{\text{liq}} - s_{\text{ice}}))^{\tau_d} D_i \quad (12)$$

Here ε_0 is the bulk porosity of CL. s_{ice} and s_{liq} are volume fractions of ice and liquid water. τ_d is the tunable parameters that affect gas transport in CL. D_i is the hydrogen and oxygen diffusivity coefficient.

Over recently studies, water in the PEM fuel cell appears as vapour, supercooled water, ice, membrane water and frozen membrane water [31,47]. During the operation of PEMFC, the product of water is supposed to be in the dissolved phase, which is considered membrane water. The liquid water desorbs from the ionomer when the membrane gradually becomes saturated in the catalyst layer. The liquid water is thermodynamically unstable, i.e., supercooled water at the subfreezing temperature. The ionomer hydration and dehydration with vapour, the evaporation and condensation between liquid water and vapour can be neglected safely as the extremely low saturated vapour at subfreezing

temperature. The supercooled water will change into ice, while Jiao [31] etc. assumed the ice crystals existed inside the ionomer called frozen membrane water. However, recent experiments [48,49] revealed that the saturated membrane water desorbed from the ionomer and froze on the ionomer surface while no ice was inside the ionomer. The frozen membrane water and ice sublimation are not considered here.

The supercooled water transport in the PEM fuel cell can be solved by

$$\frac{\partial(\varepsilon\rho_{\text{liq}}s_{\text{liq}})}{\partial t} = \frac{\partial^2(\rho_{\text{liq}}\frac{K_{\text{liq}}}{\mu_{\text{liq}}}p_c)}{\partial x^2} + S_{\text{liq}} \quad (13)$$

ε is the CL and GDL porosity while ρ_{liq} and s_{liq} are the liquid water mass density and volume fraction. K_{liq} refer to the liquid water permeability in CL and GDL described as $K_{\text{liq}} = K_0(1 - s_{\text{ice}})^4 s_{\text{liq}}^4 \mu_{\text{liq}}$ denotes the liquid water dynamical viscosity while p_c is the liquid water capillary pressure described as

$$p_c = \begin{cases} \frac{\sigma_{\text{liq}} \cos \theta}{(K_0/\varepsilon)^{0.5}} [1.417(1 - s_{\text{liq}}) - 2.12(1 - s_{\text{liq}})^2 + 1.263(1 - s_{\text{liq}})^3] & \theta \leq 90^\circ \\ \frac{\sigma_{\text{liq}} \cos \theta}{(K_0/\varepsilon)^{0.5}} [1.417s_{\text{liq}} - 2.12s_{\text{liq}}^2 + 1.263s_{\text{liq}}^3] & \theta > 90^\circ \end{cases} \quad (14)$$

θ is the contact angle relies upon the wettability of the GDL and CL. K_0 is the intrinsic permeability of the material. Here σ_{liq} represents the liquid water surface tension related to the fuel cell temperature [47].

$$\sigma_{\text{liq}} = -0.0001676 \times T + 0.1218 \quad (15)$$

The mass conservation equation of membrane water transport in the polymer membrane is solved by

$$\frac{\rho_m}{EW} \frac{\partial(\omega\lambda_m)}{\partial t} + \frac{\rho_m}{EW} \frac{\partial}{\partial x} \left(-\omega^{1.5} D_m \frac{\partial \lambda_m}{\partial x} + \frac{2.5}{22} \frac{I_{\text{ion}}}{F} \lambda_m \right) = S_{\text{mw}} \quad (16)$$

Here ρ_m is the mass density of the ionomer membrane while EW denotes the equivalent molar weight. ω is the ionomer volume fraction and S_{mw} is the membrane water source term. D_m is the diffusion coefficient of membrane water, which relies mainly upon the membrane water content and temperature.

$$D_m = \begin{cases} 3.1 \times 10^{-7} \lambda_m [\exp(0.28\lambda_m) - 1] \exp\left(\frac{-2346}{T}\right) & \text{if } 0 < \lambda_m < 3 \\ 4.17 \times 10^{-8} \lambda_m [161 \exp(-\lambda_m) + 1] \exp\left(\frac{-2346}{T}\right) & \text{if } 3 \leq \lambda_m < 17 \end{cases} \quad (17)$$

The water in the ionomer is bound to protons and transport together in the membrane. Water content represents the number of water molecules per sulfonic acid group and is defined as water molar concentration held in the membrane.

$$\lambda_m = \frac{c_{\text{mw}} EW}{\rho_{\text{mem}}} \quad (18)$$

The source terms of membrane water are defined as

$$S_{\text{mw}} = \begin{cases} -S_{m-1}, & \text{ACL} \\ 0, & \text{PEM} \\ -S_{m-1} + M_{H_2O}j/2F, & \text{CCL} \end{cases} \quad (19)$$

Here S_{i-j} represent the water phase change from i to j described in following Section 2.3.

As for the ice, the mass conservation equation is

$$\varepsilon \frac{\rho_{\text{ice}}}{M_{H_2O}} \frac{\partial s_{\text{ice}}}{\partial t} = S_{i-1} \quad (20)$$

Here ρ_{ice} is the ice mass density and M_{H_2O} is the molar mass of water. S_{i-1} represent the phase change source term of supercooled water in CL and GDL.

Table 3
Heat source.

	Reaction heat	Joule	Latent heat
PEM	0	$\sigma_e^{\text{eff}}(\nabla\phi_e)^2$	0
ACL	0	$\sigma_e^{\text{eff}}(\nabla\phi_e)^2 + \sigma_s^{\text{eff}}(\nabla\phi_s)^2$	0
CCL	$-j\frac{T\Delta S}{2F} + \eta j $	$\sigma_e^{\text{eff}}(\nabla\phi_e)^2 + \sigma_s^{\text{eff}}(\nabla\phi_s)^2$	$h_{m-1}S_{m-1} + h_{l-1}S_{l-1}$
GDL	0	$\sigma_s^{\text{eff}}(\nabla\phi_s)^2$	$h_{l-1}S_{l-1}$
B.P.	0	$\sigma_s^{\text{eff}}(\nabla\phi_s)^2$	0

The energy conservation equation of the PEM fuel cell is solved by

$$(\rho c_p)_{\text{eff}} \frac{\partial}{\partial t} T + \nabla \cdot (-k_{\text{eff}} \nabla T) = S_{\text{heat}} \quad (21)$$

Here $(\rho c_p)_{\text{eff}}$ and k_{eff} are the effective volumetric heat mass and thermal conductivity. S_{heat} is the heat source terms including reversible heat, activation heat, ohmic heat and water phase change latent heat. $\Delta S = -163.11 \text{ J/mol/K}$ is the oxygen reduction reaction entropy change. According to Thompson's study [50], the membrane water's enthalpy is the same as liquid water. All the heat source terms during the cold start process are shown in Table 3.

2.3. Water phase change model

The water inside the PEM fuel cell includes membrane water, liquid water and ice under the cold start operation. The phase change between different water states strongly influences the fuel cell cold start capacity. the oxygen reduction reaction (ORR) takes place in CCL, and water produced is desorbed by ionomer as membrane water. The ionomer membrane will desorb supercooled water and flow into CL and GDL

when the ionomer reaches saturation. The water phase change source term between membrane water and supercooled water is defined as

$$S_{m-1} = \begin{cases} \gamma_{m-1} \frac{\rho_m}{EW} (\lambda - \lambda_{\text{sat}}), & \lambda > \lambda_{\text{sat}} \& T < T_F \\ 0, & \text{others} \end{cases} \quad (22)$$

Here γ_{m-1} is the membrane water absorption and desorption rate with liquid water. s_{ice} and s_{liq} is the saturation of liquid water and ice in CL. λ_{sat} refer to the saturation membrane water content. According to the experiment data, the threshold value for the membrane's maximum water absorption varies with temperature. The maximum membrane water content λ_{sat} was obtained by the experimental test carried out by Guillermo al[51]. We refitted the experiment data with a rational polynomial with temperature:

$$\lambda_{\text{sat}} = 6.346 \times 10^{-5} T^3 - 0.04306 T^2 + 9.762 T - 731.4 \quad 223.15 \text{ K} \leq T < T_{\text{freeze}} \quad (23)$$

Here T_F refer to the freezing temperature point 273.15 K. The supercooled water is thermodynamically unstable at the subfreezing temperature and will nucleate and crystal. According to Dursch[52–54] et al. studies, ice nucleation and crystallization kinetic in the catalyst layer and GDL are obtained from isothermal differential scanning calorimetry (DSC) heat flow measurement. Then the nonlinear pseudo-isothermal ice-crystallization rate expression is derived from Johnson-Mehl-Avrami-Kolmogorov (JMAK) theory.

Under the pseudo-thermal condition, the ice-crystallization rate R_I is given by

$$R_I(T, \phi) = k(T)^{2/5} [1 - \phi] [-\ln(1 - \phi)]^{3/5} \quad \text{for } t \geq \bar{\tau}_I \quad (24)$$

Where T is average crystallization temperature, $\bar{\tau}_I$ is the number

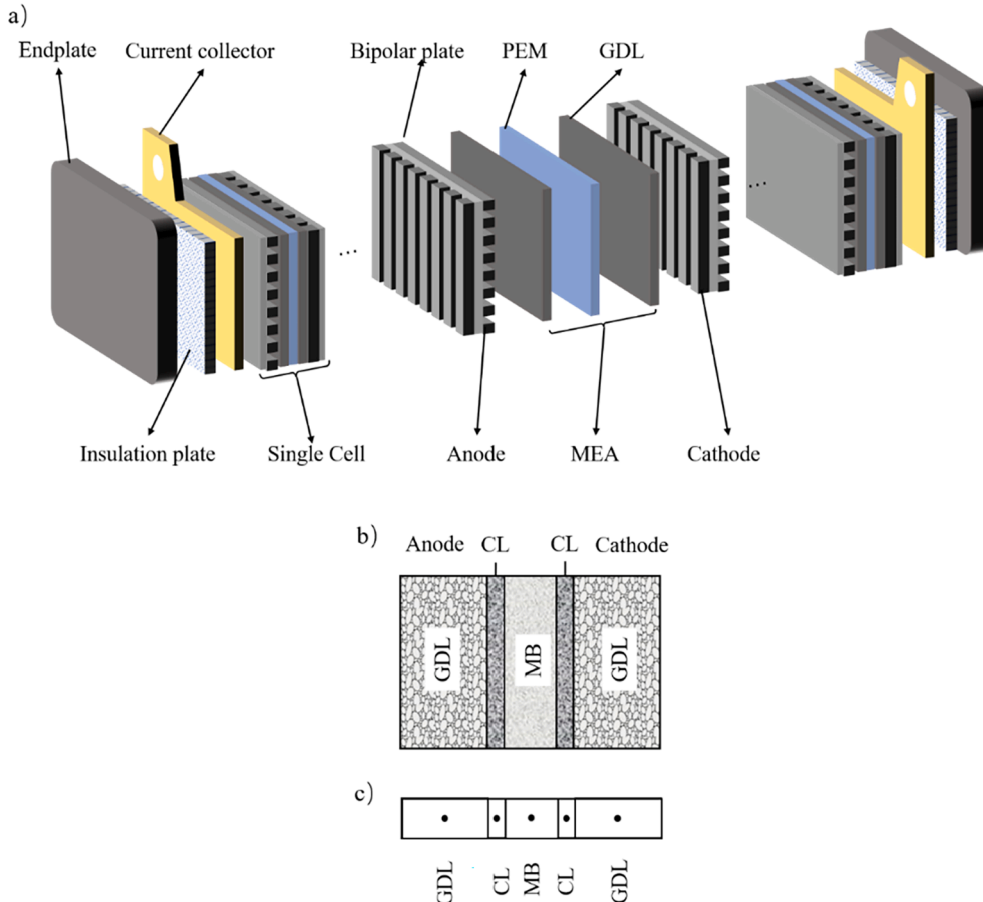


Fig. 2. A) pem fuel cell stack b) single cell modelling domain c) control volume spatial discretization.

average non-isothermal induction time, ϕ is the ice volume fraction within the pores, which is defined by $\phi = s_{ice}/(s_{ice} + s_{liq})$. $k(T)$ in Eq. is the overall ice crystallization rate constant, defined by

$$k(T) = \frac{64\pi}{15} g(\theta) J(T) \eta_o^3(T) \alpha_L^{3/2} \quad (25)$$

Here θ is the contact angle of liquid with porosity medium, $g(\theta)$ is heterogeneous nucleus growth on a flat surface, α_L is the thermal diffusivity of liquid water. $\eta_o(T)$ is a dimensionless temperature-dependence growth parameter, which is defined as

$$\eta_o(\Delta T) = \frac{a\Delta T}{1 + a\Delta T} \quad (26)$$

Here $a = 0.056K^{-1}$, ΔT is the subcooled temperature, $\Delta T = 273.15 - T$. $J(T)$ is the pseudo-steady nucleation rate of supercooled water. Which is temperature-dependence and given as

$$J(T) = A \exp \left[-\frac{B}{T(\Delta T)^2} \right] \quad (27)$$

While the Eq. only applied as time over $\bar{\tau}_i$. The non-isothermal induction time is given as

$$\int_0^{\bar{\tau}_i} \frac{dt}{\bar{\tau}_i(T)} = 1, \quad (28)$$

Where $\bar{\tau}_i(T)$ is the isothermal induction time, defined as $\bar{\tau}_i(T) = 1/J(T)V_0$, where V_0 is the liquid water volume in either CL or GDL. dt is the incremental of time interval corresponding to an isothermal induction time. Division by $\bar{\tau}_i(T)$ given the fractional contribution to $\bar{\tau}_i$ in each incremental time. When the fractional integrates to unity, the non-isothermal induction time is reached. Finally, the ice source term in CL and GDL are defined as

$$S_{l-i} = \rho_{liq}(s_{ice} + s_{liq})R_i \quad (29)$$

3. Model order reduction and simplification

Although the one-dimensional cold-start model is more straightforward than a three-dimensional model, the distributed model is still computationally costly. In order to use the cold start model in the control system, model order reduction and simplification are required. McKay [40] et al. discrete the GDL into three parts when reducing the two-phase flow model at room temperature. Each part of the sub-volume is supposed to be homogenous. Then the PDE equations of the Multiphysics process are simplified into ordinary differential equations. However, the catalyst layer is neglected safely because the liquid water in CL is much less than in GDL. However, during a cold start, the electrochemical reaction rate in the CL is serious inconsistency, and the ice formation in the catalyst layer is strongly uneven. The species transport process in the catalyst layer is essential, mainly affecting the cold start performance. Jiang [43] et al. et al. realize model order reduction by control volume method. The one-dimensional model has been split into twelve parts, while the catalyst layer is divided into two-part to investigate nonuniformity distribution. Nevertheless, the simplification model is still too complicated to integrate into system control.

In this section, to simplify the 1-D cold-start model, the distributed model is divided into five regions, and the catalyst layer is considered, as Fig. 2 shows. The Gas Channel, GDL, CL and polymer membrane are five regions to describe the Multiphysics process. The exact model reduction method of each sub-volume is introduced in the following Sections.

3.1. Gas species transport

The reaction species transport can be considered a steady-state because gas species diffuse much faster than water transport in the ionomer membrane. Diffusive transport is only considered in the porous

media domain, and convective transport is neglected, mainly in the gas channel. For all the reaction species concentrations in the model domains, the concentrations in the CL are mainly considered, affecting the output voltage through the electrochemical dynamic process. Furthermore, reaction species concentrations are determined by the gas channel supplied species concentration and the reaction molar flux, calculated by the electrochemical reaction rate at the catalyst layer, just as the local current density. Here the molar flux of reaction species is as follows $N_{O_2} = \frac{I_{dens}}{4F}$ and $N_{H_2} = \frac{I_{dens}}{2F}$. The reaction concentration at the active sites is calculated as:

$$c_{O_2,CL} = c_{O_2,CH} - R_{O_2} \frac{I_{dens}}{4F} \quad (30)$$

$$c_{H_2,CL} = c_{H_2,CH} - R_{H_2} \frac{I_{dens}}{2F} \quad (31)$$

Here $c_{O_2,CH}$ and $c_{H_2,CH}$ are the concentration of oxygen and hydrogen in the gas channel. R_{O_2} and R_{H_2} are the species transport resistance from gas channel to reaction site. Which are defined as:

$$R_i = R_{i,CH} + R_{i,GDL} + R_{i,CL} \quad (32)$$

The first term on the right hand of Eq. describes the convective transport resistance at the gas channel and GDL interface, described by the Sherwood number:

$$R_{i,CH} = \frac{D_h}{Sh \cdot D_i} \quad (33)$$

Where D_h is the hydraulic diameter of gas channel. Sh is the Sherwood number, while D_i is the reaction species diffusive coefficient. $R_{i,GDL}$ and $R_{i,CL}$ are the diffusive resistance through the GDL and CL, which are calculated by the species diffusive coefficient and porous media porosity:

$$R_{i,layer} = \frac{\delta_{layer}}{D_i^{eff}} \quad (34)$$

Ice cannot completely fill the electrode pore volume for higher current and progressively lower temperatures, mainly due to unevenly current density distribution in the CL. In order to solve this challenging problem, here we introduce a parameter, CL pore volume utilization rate, $USED$, which is a fitting parameter and defined as [55]

$$\%USED = \begin{cases} -0.2364 \ln(i) + 0.0235i \leq 300 \\ 0.165i > 300 \end{cases} \quad (35)$$

Here i is the current density, $USED$ is mainly employed to describe the oxygen transport resistance in CL.

3.2. Membrane water

Three regions are chosen to describe the water transport behaviour while the ionomer exists in the catalyst layer and the middle polymer membrane. The water in the ionomer transport is dominated by two mechanisms, diffusion and electro-osmotic drag. Then the membrane water content is defined as:

$$\omega_{CL}^{an} \frac{d\lambda_{CL}^{an}}{dt} = \frac{J_{acl-m}^d - J_{acl-m}^e}{\delta_{CL}} \quad (36)$$

$$\frac{d\lambda_m}{dt} = \frac{J_{ccl-m}^d - J_{acl-m}^d + J_{acl-m}^e - J_{ccl-m}^e}{\delta_m} \quad (37)$$

$$\omega_{ion}^{ca} \frac{d\lambda_{CL}^{ca}}{dt} = \frac{1}{\delta_{CL}} (J_{ccl-m}^e - J_{ccl-m}^d + \frac{I_{dens}}{2F}) - S_{ml} \quad (38)$$

Where λ_{CL}^{an} , λ_m and λ_{CL}^{ca} are the membrane water content in the anode CL and PEM and cathode CL. J_i^d and J_i^e represent the diffusion flux and electro-osmotic drag flux. While the volume fraction of the ionomer is discontinuous at the CL and PEM interface, the diffusivity fluxes at the

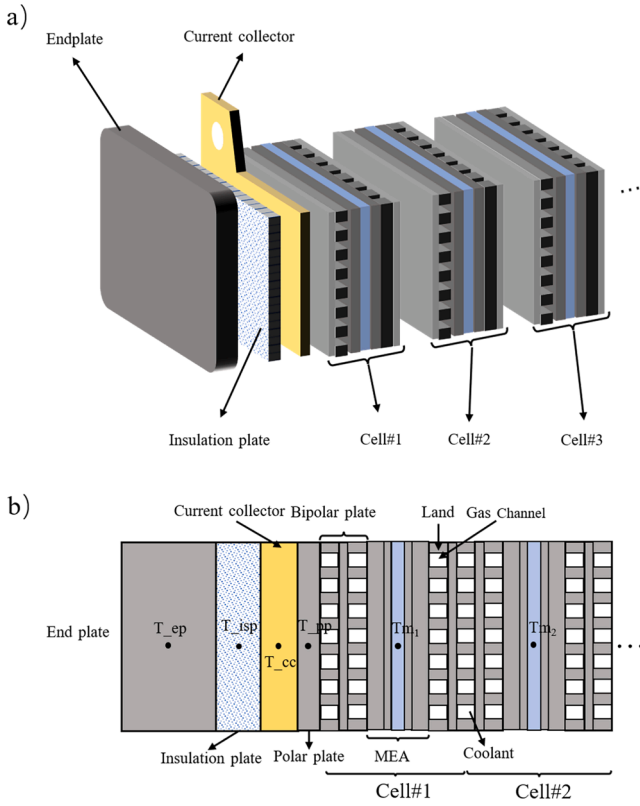


Fig. 3. A) pem fuel cell stack b) pem fuel cell stack heat modelling domain.

interface are described as

$$J_{acl-m}^d = \omega_{CL}^{1.5} K_d D_{acl-m} \frac{(\lambda_{acl-m} - \lambda_{CL}^{an})}{0.5\delta_{CL}} = K_d D_{acl-m} \frac{(\lambda_m - \lambda_{acl-m})}{0.5\delta_m} \quad (39)$$

$$J_{acl-m}^e = \frac{2.5}{22} \frac{I_{dens}}{F} \lambda_{acl-m} \quad (40)$$

$$J_{ccl-m}^d = \omega_{CL}^{1.5} K_d D_{ccl-m} \frac{(\lambda_{ca}^{ca} - \lambda_{ccl-m})}{0.5\delta_{CL}} = K_d D_{ccl-m} \frac{(\lambda_{ca}^{ca} - \lambda_m)}{0.5\delta_m} \quad (41)$$

$$J_{ccl-m}^e = \frac{2.5}{22} \frac{I_{dens}}{F} \lambda_{ccl-m} \quad (42)$$

Where K_d is the corrected parameter of membrane water diffusivity coefficient, which needs to be identified in Section 4. The distribution of membrane water is not linear. The interface membrane water content can be calculated by interface diffusivity flux of CL and membrane water content:

$$\lambda_{acl-m} = \left(\frac{\omega_{an}^{1.5} \lambda_{CL}^{an}}{\delta_{CL}} + \frac{\lambda_m}{\delta_m} \right) / \left(\frac{\omega_{an}^{1.5}}{\delta_{CL}} + \frac{1}{\delta_m} \right) \quad (43)$$

$$\lambda_{ccl-m} = \left(\frac{\omega_{ca}^{1.5} \lambda_{CL}^{ca}}{\delta_{CL}} + \frac{\lambda_m}{\delta_m} \right) / \left(\frac{\omega_{ca}^{1.5}}{\delta_{CL}} + \frac{1}{\delta_m} \right) \quad (44)$$

3.3. Liquid water transport

When ionomer in the catalyst layer uptake water and reach saturated, excess water will desorb from the membrane as supercooled water in the liquid state. The liquid water fills the pore spaces in the catalyst layer and flows to the adjacent pore in GDL during the capillary pressure increase. Most unstable supercooled water will nucleate and crystallize, turning into ice. The result of the liquid water will transport across the CL and GDL, resulting in an injection of liquid water into the channel.

Here we suppose the liquid water in the channel is mist flow, the liquid water saturation at the boundary of the channel, and GDL is zero. Applying the mass conservation equation to CL and GDL, the liquid water saturation is calculated as:

$$\begin{cases} \epsilon_{CL} \frac{\rho_{lq}}{M_{H_2O}} \frac{ds_{lq}^{cl}}{dt} = \epsilon_{CL} \frac{\rho_{lq}}{M_{H_2O}} \frac{-W_{cl2gdl}}{\delta_{CL}} + S_{ml} - S_{li} \\ \epsilon_{GDL} \frac{\rho_{lq}}{M_{H_2O}} \frac{ds_{lq}^{gdl}}{dt} = \epsilon_{GDL} \frac{\rho_{lq}}{M_{H_2O}} \frac{W_{cl2gdl} - W_{gdl2ch}}{\delta_{GDL}} - S_{li} \end{cases} \quad (45)$$

Where s_{lq}^{cl} and s_{lq}^{gdl} are the volume fraction of liquid water in CL and GDL. S_{ml} and S_{li} are the liquid water source terms, S_{ml} is the liquid water desorbed from ionomer water, S_{li} describe the liquid change into ice. W_{cl2gdl} is the flux of liquid water from CL to GDL while W_{gdl2ch} represent liquid water transport from GDL to gas channel. The flux of liquid water transport is defined as:

$$\begin{cases} W_{cl2gdl} = -\frac{\rho_l K_{lq}}{\epsilon_{cl} \mu_{lq}} \left(\frac{dp_c}{ds_{lq}^{cl}} \right) \frac{s_{lq}^{cl} - s_{lq}^{gdl}}{0.5(\delta_{cl} + \delta_{gdl})} \\ W_{gdl2ch} = -\frac{\rho_l K_{lq}}{\epsilon_{gdl} \mu_{lq}} \left(\frac{dp_c}{ds_{lq}^{gdl}} \right) \frac{s_{lq}^{gdl}}{0.5\delta_{gdl}} \end{cases} \quad (46)$$

Here p_c and K_{lq} are the capillary pressure and effective permeability of liquid water transport defined in Section 2.2.

3.4. Heat transport

PEM fuel cell stack cold start process follows the cannikin's law. The end cell's cold start performance determines whether the fuel cell stack can be cold-started successfully. Meanwhile, the endplate's thermal mass effect the end cell's temperature distribution. Therefore, a lumped stack heat model cannot capture the temperature distribution across the fuel cell stack thermal condition. Model considering individual cells can reveal the effect of the current collectors and endplate thermal mass on the end cell's temperature. Then the temperature of the individual fuel cell heat model is developed here. The PEM fuel cell stack and model heat transport domain are shown in Fig. 3.

The energy conservation equation calculates the temperature during the cold start for individual cells.

$$C_p^{eff} \frac{dT_{mi}}{dt} = S_{heat} \quad (47)$$

Here C_p^{eff} is the individual cell's effective thermal mass, including MEA, bipolar plates and coolant. The reaction species and ice influence on heat transfer are neglected to simplify the calculation. Meanwhile, the insulation plate is made of FR4 material, and the heat transfer coefficient is much tiny, about 0.25 W/m/K. Then the heat is brutal to transfer to the endplate, so we neglect endplate thermal mass influence. The temperature at the endplate and insulation plate interface is the same as the ambient temperature.

The individual cell's effective thermal mass is defined as

$$C_p^{eff} = A_{fc} (c_{p,m}^{eff} \delta_m + 2c_{p,cl}^{eff} \delta_{cl} + 2c_{p,gdl}^{eff} \delta_{gdl} + 2c_{p,bp}^{eff} \delta_{bp}) \quad (48)$$

The effective thermal mass of each part in the PEM fuel cell is calculated

$$\begin{cases} c_{p,m}^{eff} = \rho_m c_{p,m} \\ c_{p,cl}^{eff} = \omega \rho_m c_{p,m} + (1 - \epsilon_{cl} - \omega) \rho_{cl} c_{p,cl} \\ c_{p,gdl}^{eff} = (1 - \epsilon_{gdl}) \rho_{gdl} c_{p,gdl} \\ c_{p,bp}^{eff} = \rho_{bp} c_{p,bp} \frac{\delta_{bp} A_{fc} - V_{gas} - V_{coolant}}{\delta_{bp} A_{fc}} + \rho_{coolant} c_{p,coolant} \frac{V_{coolant}}{\delta_{bp} A_{fc}} \end{cases} \quad (49)$$

Here V_{gas} and $V_{coolant}$ are separately the volume of the gas channel and coolant channel. The ohmic heat in the bipolar is negligible for the high

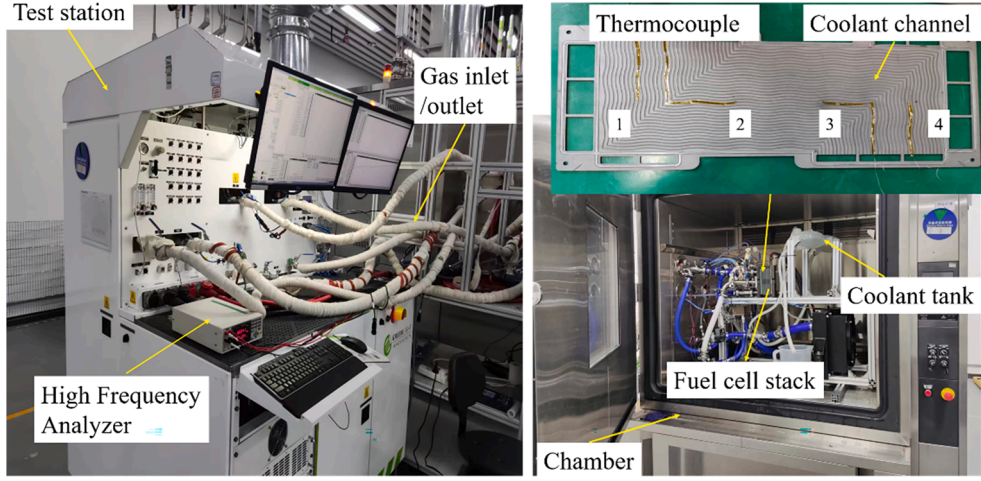


Fig. 4. Fuel cell cold start Experimental conditions.

Table 4
Identification parameters.

Parameter[units]	Range	Experimental identified value
$K_d[-]$	[0.1,10]	3.23
$\gamma_{ml}[1/s]$	[0.1,10]	1.2
$E_i[J/mol]$	[50000,100000]	60,485
$A_T[nuclei/m^3/s]$	$[1,100] \times 10^8$	19.34×10^8
$B_T[K^3]$	$[1,50] \times 10^4$	13.13×10^4
τ_a	[1,20]	19.9
τ_d	[1,5]	3.72

conductivity. Other heat sources are calculated as

$$\left\{ \begin{array}{l} S_{heat}^{gdl} = \frac{2I_{cell}^2 H_{gdl}}{\sigma_{gdl}^{eff}} \\ S_{heat,acl} = I_{cell}^2 R_{CL}^{an} \\ S_{heat,pem} = I_{cell}^2 R_m \\ S_{heat,ccl} = I_{cell}^2 R_{CL}^{ca} + \left| I_{cell} \frac{T \Delta S}{2F} \right| + I_{cell} |\eta| - h_{desorb} S_{ml} + h_{fusion} S_{li} \\ S_{heat,exchange} = A_{fc} k_{m_{i+1}-m_i}^{eff} \frac{T_{m_{i+1}-m_i} - T_{m_i}}{0.5\delta_{fc}} + A_{fc} k_{m_{i-1}-m_i}^{eff} \frac{T_{m_{i-1}-m_i} - T_{m_i}}{0.5\delta_{fc}} \end{array} \right. \quad (50)$$

Here R_{CL}^{an} and R_{CL}^{ca} are separately the anode and cathode catalyst layer resistance, while R_m is the membrane resistance. All of the resistances are defined in section 3.6. The detail of heat sources during the cold start shown in table III. $S_{heat,exchange}$ is heat exchange flow with neighbour cell. While δ_{fc} is defined as $\delta_{fc} = 0.5\delta_m + \delta_{cl} + \delta_{gdl} + \delta_{bp}$, A_{fc} is the fuel cell effective reaction area. $k_{m_{i+1}-m_i}^{eff}$ is the effective thermal conductivity coefficient between cell# i and cell# $i+1$. Here we take the bipolar plate as a uniform structure, and the interface's effective thermal conductivity is consistent with the thermal conductivity of the bipolar plate k_{bp}^{eff} . Bipolar plate effective thermal conductivity is defined as

$$k_{bp}^{eff} = k_{bp} \frac{\delta_{bp} A_{fc} - V_{gas} - V_{coolant}}{\delta_{bp} A_{fc} - V_{gas}} + k_{coolant} \frac{V_{coolant}}{\delta_{bp} A_{fc} - V_{gas}} \quad (51)$$

While fuel cell stack cold start performance is determined by the end cell. The heat transfer between cell#2 and the polar plate coincides with the end cell#1 cold start process. The boundary condition between the insulation and end plates is set as ambient temperature $T_{isp-ep} = T_{amb}$. The end plate assembly includes an insulation plate, current collector, and polar plate with no inner heat source. The heat exchange flux

between neighbour components is defined as

$$\left\{ \begin{array}{l} S_{heat,exchange}^{pp-m1} = k_{bp}^{eff} \frac{T_{m1} - T_{m1-pp}}{0.5\delta_{fc}} = k_{pp} \frac{T_{m1-pp} - T_{pp}}{0.5\delta_{pp}} \\ S_{heat,exchange}^{pp-cc} = k_{pp} \frac{T_{pp} - T_{pp-cc}}{0.5\delta_{pp}} = k_{cc} \frac{T_{pp-cc} - T_{cc}}{0.5\delta_{cc}} \\ S_{heat,exchange}^{cc-isp} = k_{cc} \frac{T_{cc} - T_{cc-isp}}{0.5\delta_{cc}} = k_{isp} \frac{T_{cc-isp} - T_{isp}}{0.5\delta_{isp}} \\ S_{heat,exchange}^{isp-amb} = k_{isp} \frac{T_{isp} - T_{amb}}{0.5\delta_{isp}} \end{array} \right. \quad (52)$$

Here T_{i-j} is the interface temperature between components i and j . The insulation plate is made of well thermal and electric insulation material FR4. Much little heat can transfer from the insulation plate to the end plate. Here we can neglect the endplate thermal mass effect safely.

3.5. Gas channel

A lumped gas channel model is used at the GDL and B.P. boundary. The mass conservation of gas species in the cathode gas channel is defined as:

$$\left\{ \begin{array}{l} \frac{dm_{O_2, ch}}{dt} = W_{O_2, ca, in} - W_{O_2, ca, out} - W_{O_2, ca, GDL} \\ \frac{dm_{N_2, ca, ch}}{dt} = W_{N_2, ca, in} - W_{N_2, ca, out} \end{array} \right. \quad (53)$$

The cathode channel pressure is calculated $p_{ca, ch} = p_{O_2, ch} + p_{N_2, ca, ch}$. Note that the partial pressures for oxygen $p_{O_2, ch} = \frac{RT}{M_{O_2} V_{ca}} m_{O_2, ch}$, and nitrogen $p_{N_2, ch} = \frac{RT}{M_{N_2} V_{ca}} m_{N_2, ch}$.

All of the reaction species mas flows supplied to cathode gas channel are defined as

$$\left\{ \begin{array}{l} W_{O_2, ca, in} = \alpha_{O_2} \frac{M_{O_2} I_{cell} A_{fc}}{4F} \\ W_{N_2, ca, in} = \alpha_{O_2} \frac{M_{N_2} I_{cell} A_{fc}}{4F} \frac{1 - x_{O_2}}{x_{O_2}} \end{array} \right. \quad (54)$$

The mass flows of individual gas exiting the cathode channel are expressed

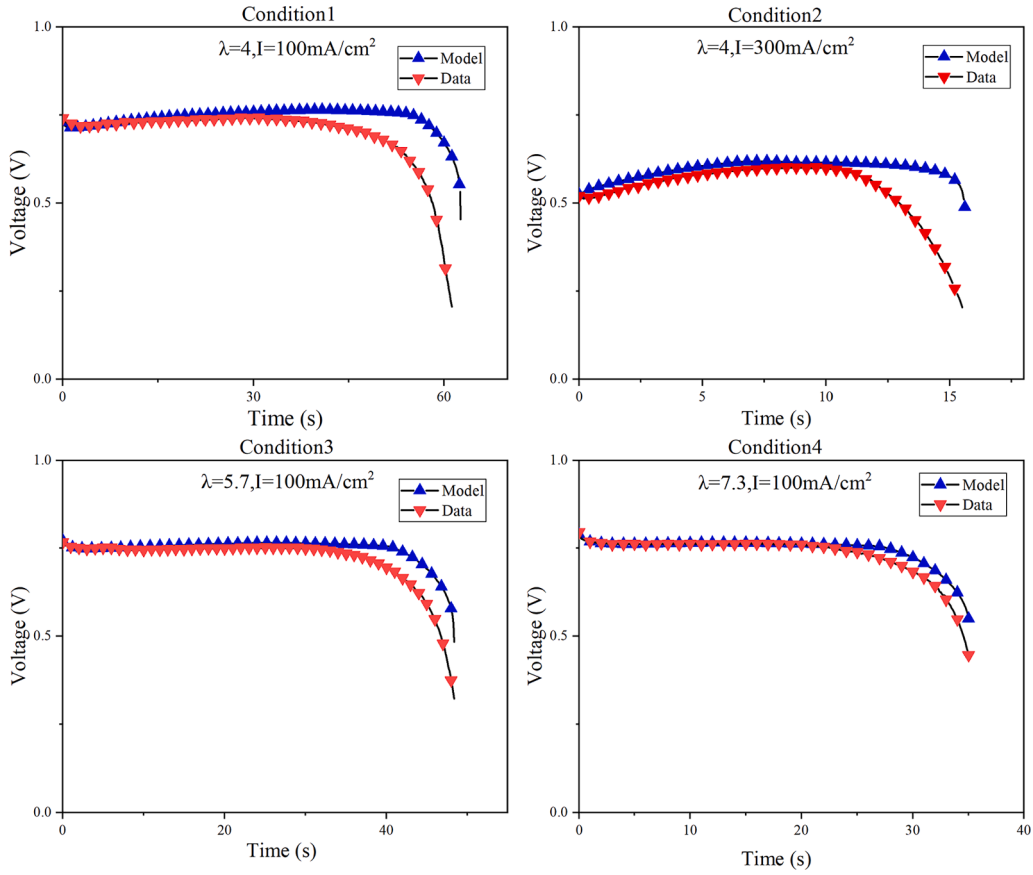


Fig. 5. Comparison of model voltage with experiment data for isothermal single cell cold start.

$$\begin{cases} W_{ca,out} = k_{ca}(p_{ca,ch} - p_{ca,out}) \\ W_{O_2,ca,out} = \frac{m_{O_2,ca,ch}}{m_{ca,ch}} W_{ca,out} \\ W_{N_2,ca,out} = \frac{m_{N_2,ca,ch}}{m_{ca,ch}} W_{ca,out} \end{cases} \quad (55)$$

k_{ca} is the orifice constant of the cathode throttle valve, which is determined by experiment measurement. $m_{ca,ch} = m_{O_2,ca,ch} + m_{N_2,ca,ch}$ is the total gas mass in the cathode gas channel. Then the mass flux that diffuses into GDL is defined as $W_{O_2,ca,GDL} = M_{O_2} \frac{I_{cell}}{4F}$. In contrast, hydrogen electrochemical reaction rate orders magnitude much faster than oxygen reaction. So, the overpotential hydrogen reaction at the anode catalyst layer is minimal. Meanwhile, the electrical drag is mainly in the membrane, and liquid water and ice mainly emerge in the cathode of the PEM fuel cell. Therefore, the hydrogen concentration in the gas channel is not considered here.

3.6. Terminal voltage and reaction kinetics

The hydrogen reaction rate at the anode catalyst is much faster than the oxygen reduction reaction at the cathode. So, the anode overvoltage is also much lower than cathode, which can be neglected safely. The terminal voltage of the fuel cell can be expressed as:

$$E_{cell} = E_{nernst} - \eta_{ORR} - I_{dens}(R_{mb} + R_{CL}^{ca,eff} + R_{CL}^{an,eff} + R_{cont}) \quad (56)$$

In the above equation, the first terms on the right hand represented the open-circuit voltage, described in Eq. The ORR kinetic model used in this work is based on the modified Tafel formula, which accounts for the overvoltage loss led by the reaction species concentration and electrochemical active surface area (ESCA) effect by liquid water and ice saturation.

$$\eta_{ORR} = \frac{RT}{\alpha_{ca}F} \left[\ln \left(\frac{I_{dens}}{i_{0,ca} \delta_{CL}^{ca} a_{Pt}^{ca}} \right) \right] \quad (57)$$

Where $i_{0,ca}$ is the exchange current density, which is modified by the reaction species concentration, liquid and ice saturation and temperature, which is described as

$$i_{0,ca} = (1 - s_{ice} - s_{liq})^{\tau_a} i_{0,ref}^{ca} \left(\frac{c_{O_2,ca}}{c_{O_2,ref}} \right) \exp \left[-\frac{E_c}{RT} \left(1 - \frac{T}{T_0} \right) \right] \quad (58)$$

E_c the activation energy for oxygen reduction reaction, which needs to be tuned. T_0 is the reference temperature and $c_{O_2,ca}$ is the cathode CL oxygen concentration. τ_a is the electrochemical active surface area corrected parameter affected by the ice and liquid water. τ_a is strongly dependent on the microscale monopoly of ice in CL. According to Mao etc. [56,57] studies, if the ice formed in CL is a sheet-like shape, the reaction surface would be covered by a small volume fraction of ice, and the corrected parameter will be relatively large. If the ice morphology is needlelike, the exponent τ_a will be exceedingly small. Here the τ_a is constant at 1 when the ice saturation is small than 0.3. When the ice volume fraction exceeds 0.3, τ_a increase significantly until the cold start fails.

$$\tau_a = \begin{cases} 1 & s_{ice} \leq 0.3 \\ 1 + \frac{s_{ice} - 0.3}{0.7} \cdot \tau_a s_{ice} & s_{ice} > 0.3 \end{cases} \quad (59)$$

The ohmic voltage loss is mainly dominated by the membrane conductivity, catalyst layer ohmic resistance, and bulk electrical resistance of conductive material. According to Fang [39], the membrane resistance is calculated by two linear distributions of membrane water and proton conductivity.

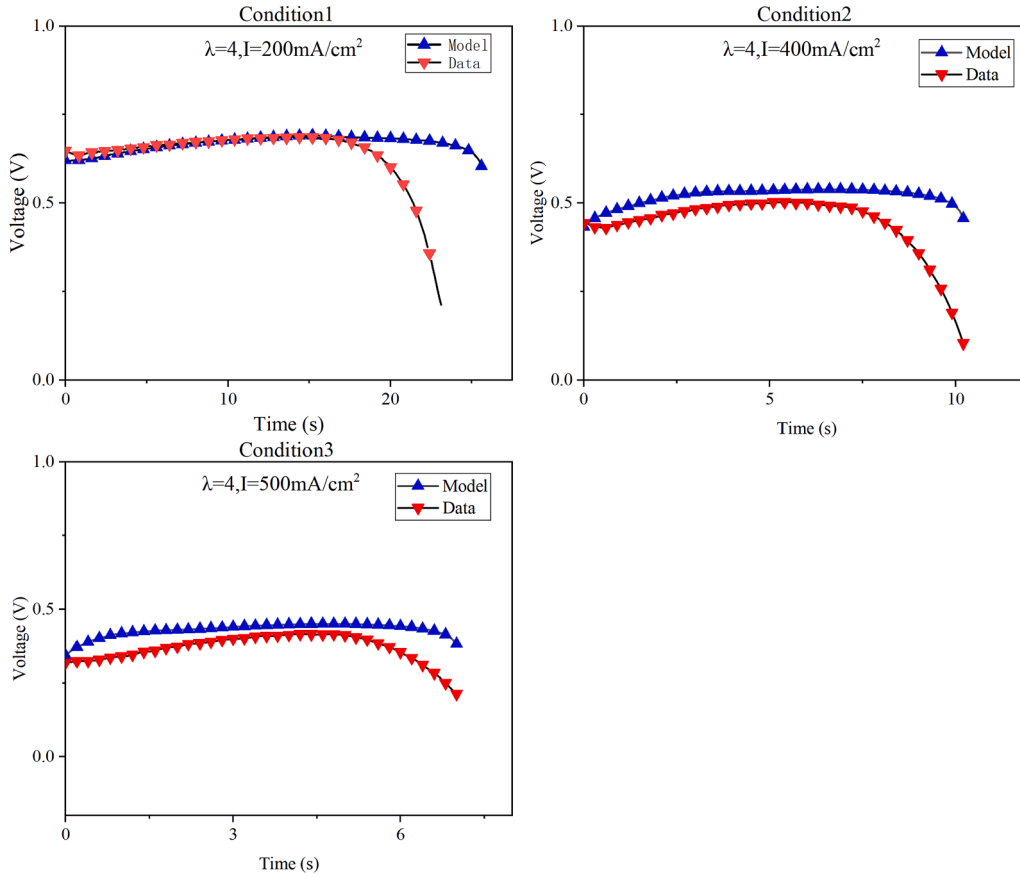


Fig. 6. Model validation results for isothermal cold start at large current density.

$$R_m = \frac{\delta_{mem}}{2(\sigma_m - \sigma_{acl-m})} (\ln \sigma_m - \ln \sigma_{acl-m}) + \frac{\delta_{mem}}{2(\sigma_{ccl-m} - \sigma_m)} (\ln \sigma_{ccl-m} - \ln \sigma_m) \quad (60)$$

σ_{acl-m} and σ_{ccl-m} are the proton conductivity of the membrane at the interface of the PEM and catalyst layer. Meanwhile, σ_m is the proton conductivity in the middle of membrane, which is firmly dependent on the membrane water content λ_{acl-m} , λ_{ccl-m} and λ_m . Where λ_{acl-m} and λ_{ccl-m} are the membrane water content at the interface of the membrane and catalyst layer. Catalyst layer resistance includes electrode material resistance and ionomer resistance. Due to the current density distribution nonuniform at the catalyst layer [10,43,58], the effective electrode resistance is modified as

$$R_{CL}^{ca,eff} = \frac{\delta_{cl}}{(1 - \epsilon_{cl} - \omega_{cl})^{1.5} \sigma_{ele,cl}} + \frac{\delta_{cl}}{(3 + \zeta) \sigma_{eff}} \quad (61)$$

The first term on the right hand is the bulk electron conductivity material resistance. At the same time, the second term is the cathode proton transport resistance. Here ζ is the corrected catalyst utilization parameter, which requires calculating the dimensionless parameter iR_{sheet}/b , where b is the Tafel slope. b is strongly dependent on the temperature, defined as $b = 2.3 \frac{RT}{\alpha_c F}$. R_{sheet} is the ionomer proton transport resistance in the cathode catalyst layer, calculated as $\frac{\delta_{cl}}{\omega^{1.5} \sigma_m}$.

$$\zeta = 5.1952e - 5 * (iR_{sheet}/b)^3 - 6.156e - 3 * (iR_{sheet}/b)^2 + 0.68777 * (iR_{sheet}/b) \quad (62)$$

Anode catalyst layer resistance cannot be neglected. During the cold start process, the load current at the beginning of the cold start is relatively large to generate immense heat to warm up the fuel cell. A large current will decrease ionomer water content at the anode catalyst, and proton conductivity will decrease significantly. Therefore, the anode

catalyst layer resistance is calculated as [10]

$$R_{CL}^{an,eff} = \frac{\delta_{cl}}{(1 - \epsilon_{cl} - \omega_{cl})^{1.5} \sigma_{ele,cl}} + R_H^{eff} \quad (63)$$

$$R_H^{eff} = \left(\frac{e^{s\delta_{cl}} + e^{-s\delta_{cl}}}{e^{s\delta_{cl}} - e^{-s\delta_{cl}}} - \frac{1}{s\delta_{cl}} \right) / s\delta_{cl} \cdot \frac{\delta_{cl}}{\sigma_{eff}} \quad (64)$$

s is the corrected parameters $s = \left[\frac{A_{Pt} L_{Pt} i (a_a + a_c) F}{\sigma_{eff} \delta_{cl} RT} \right]^{1/2}$, representing the ratio of the effective proton conduction resistance in the anode electrode over the charge transfer resistance. σ_{eff} is the effective conductivity, described as Eq. A_{Pt} is the specific surface area of the Pt catalyst. L_{Pt} is the Pt loading of the electrode, i is the exchange current density of HOR. a_a and a_c are the charge transfer coefficients of HOR.

4. Parameter identification

4.1. Experimental data

The model is validated against PEMFC isothermal and non-isothermal cold start dates. The isothermal cold start experiment is conducted on a single PEM fuel cell with a 25 cm² active area. In contrast, the non-isothermal cold start is conducted on a 20 cells PEM fuel cell stack with a 330 cm² active area. However, the two PEM fuel cells have the same MEA, which are provided by Gore. The detail of the operation condition and high-level description of each cold start experiment are presented below. The cold start experimental condition are shown in Fig. 4. The test station is the cold start control equipment, which provides the reaction gas and records the experiment data. Meanwhile, the chamber is the environmental equipment, which provides the constant temperature for the fuel cell. High frequency analyzer is used to monitor the fuel cell's membrane condition during the purge

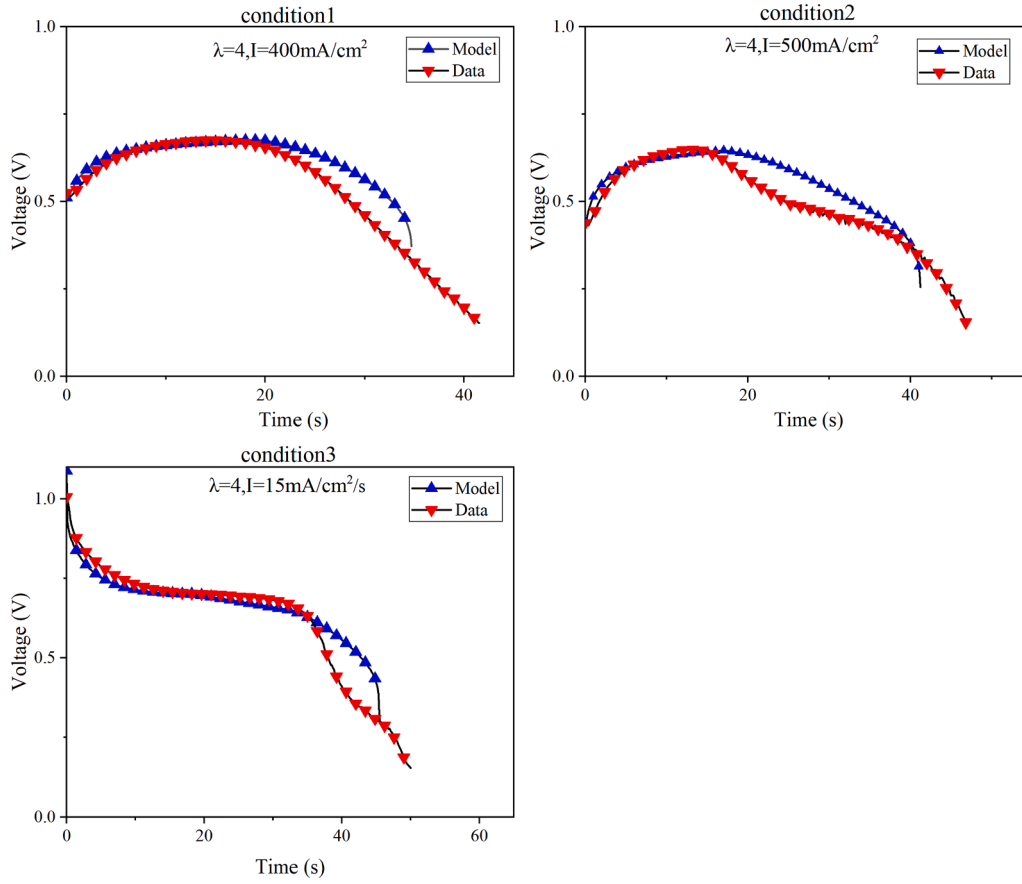


Fig. 7. Comparison of model voltage with experiment data for stack non-isothermal cold start.

and freezing process. The thermocouples are displayed in the coolant channel of end cell#1 to record the temperature variance during the cold start process.

The isothermal cold start experiment is conducted on a single cell, while the bipolar has a colossal heat mass for the single PEM fuel cell. Compared to the heat produced, the temperature rise can neglect safely. The experimental results include four different operating conditions, and initial conditions constitute a relatively wide operating range, covering current densities of 100 and 300 mA/cm² from low to high. Notably, the initial membrane water content heavily influences the cold start capability. Therefore, different initial water contents, 5.7 and 7.3, from low to high, are achieved in the isothermal cold start experiment. Model validation is carried out using a single cell's four different cold start voltage curves.

Non-isothermal cold start experiments contain voltage dataset and temperature dataset from a 20 cells size stack with 0.35 mg/cm² Pt loading at the cathode catalyst layer and a large active area (about 330 cm²). It should be noted that the material of ionomer membrane, GDL, and CL morphology used in the stack is considerably the same as a single cell. Additionally, different loading currents are used here to extensively explore the effect on cold start capacity. Constant current density, 400 mA/cm² and 500 mA/cm² are applied to the stack to cold start from −20 °C. Meanwhile, the voltage and temperature variations are recorded during the cold start. A ramping current density of 15 mA/cm²/s is

applied to investigate the dynamic current's influence on the cold start performance. Three working condition experiment datasets are used here to validate and identify the parameter of the non-isothermal cold-start model.

4.2. Model parameter identification

The isothermal model and non-isothermal model parameters related to fuel cell components are listed in Table I. The experiment data above needs to identify a set of five parameters. These identification parameters within a specific range are provided in Table 4. The range of identified parameters is determined by relevant values reported in previous literature. The first and second identified parameter is related to ionomer membrane property. K_d is used to correct the membrane diffusivity coefficient, while the tunable parameter γ_{ml} is the membrane water desorbing rate when the ionomer reaches saturation. Here E_c is the activation energy used to determine the current exchange density at different temperatures. The last two tunable parameters A_T and B_T are related to the freezing water rate. τ_a and τ_d are the corrected parameters of liquid water and ice saturation affected the ECSA and gas transport in CL. Afterwards, identified parameters values are explored to minimize the following cost function in the specified range:

$$J = \sum_{\text{Isothermal}} \frac{\|[\bar{v}(\tau) - \hat{v}_I(\tau)]^T [\bar{v}(\tau) - \hat{v}_I(\tau)]\|_2}{n_V} + \sum_{\text{Nonisothermal}} \frac{\|[\bar{v}(\tau) - \hat{v}_I(\tau)]^T [\bar{v}(\tau) - \hat{v}_I(\tau)]\|_2}{n_V} \quad (65)$$

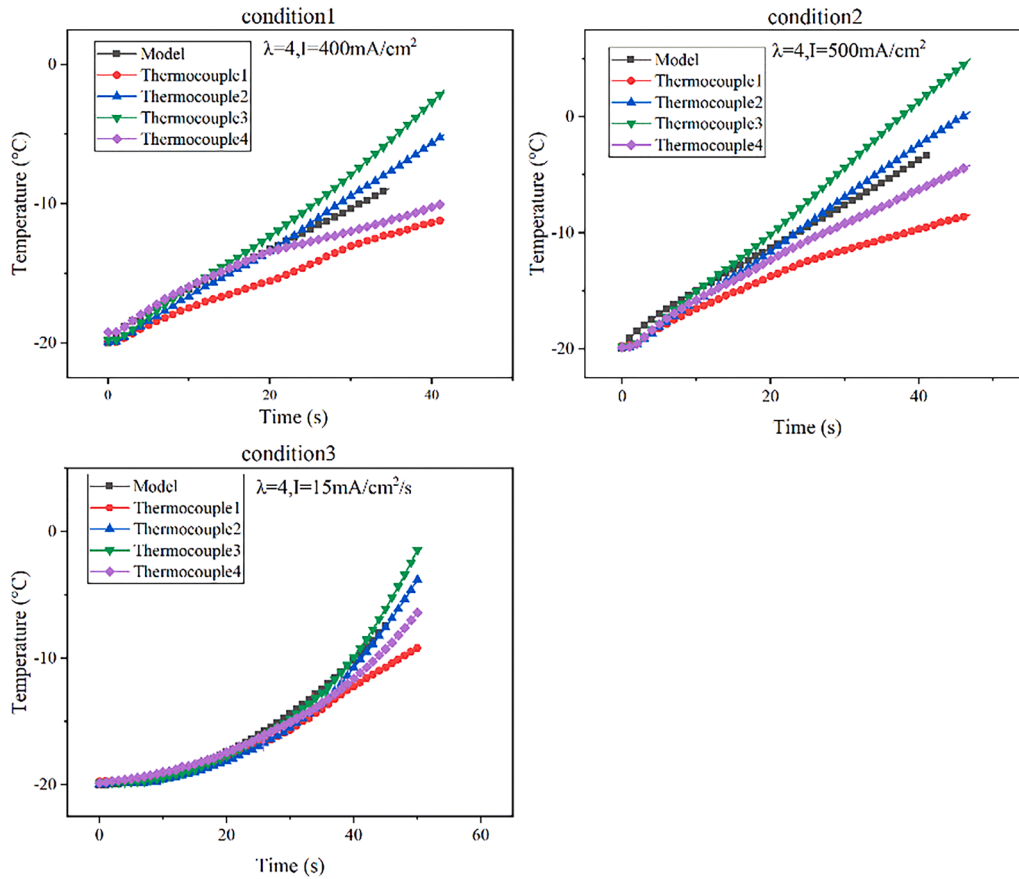


Fig. 8. Comparison of model temperature with experiment data for stack non-isothermal cold start.

Here the first term of the cost function is the sum of isothermal cold start error, including four isothermal cold start conditions. The second term is the sum of non-isothermal cold start error. $\bar{v}(\tau)$ is the model predicting the cold start voltage, while $\hat{v}_i(\tau)$ the experiment records voltages. n_v is the experiment record voltage number, and the experiment record frequency is 10 Hz.

The identification process is implemented by particle swarm optimization (PSO)[59] algorithm in the MATLAB software environment. PSO is a global optimization method exploring the whole parameter space to search for the optimal set. The swarm size is set at 100, and the iteration time is 100. When the iteration times are reached, the cost function value is not reduced anymore. The optimization algorithm is carried out for several hours to find the optimal parameter.

5. Model validation and discussion

5.1. Result for isothermal single cell cold start

The comparison of the experimental and model prediction voltage for the four cold start condition are shown in Fig. 5. Operation condition (1) to condition (4) are constant current cold start modes, which are separately initial membrane water content 4, current density 100 mA/cm²; initial membrane water content 4, current density 300 mA/cm²; initial membrane water content 5.7, current density 100 mA/cm²; initial membrane water content 7.3, current density 100 mA/cm². Overall, the model prediction voltage can well capture the output voltage trend in experimental data.

As we can see, the higher the initial membrane water content, the less survival time the cold start process. The higher initial membrane water content will make the ionomer easily reach saturation, saturation liquid water release from ionomer to porosity and freezing to ice and the

cold start process terminates. Meanwhile, higher load current density will result in a significant water production rate, which also causes cold start to fail quickly. As Fig. 5. shows, the cold start voltage can increase at the initial stage, and the ionomer membrane uptake water improves membrane conductivity. Therefore, the ohmic membrane resistance decreases, increasing the output voltage. The saturation liquid water freezes to ice at the voltage failure stage and exists in the catalyst layer. The ice impedes oxygen transport to Pt/C particle reaction while the ECSA also decreases due to the ice deposition. The activation over-voltage increases significantly by oxygen concentration, and ECSA decreases. As Fig. 5 shows, the model prediction voltage and the experiment voltage are a little inconsistent at the voltage failure stage. The error is mainly due to the influence of ice morphology on oxygen concentration and ice volume fraction.

In contrast, the existence error has little influence on the model prediction cold start output voltage and survival time. According to the error analysis, the average of the four conditions is 0.8 %,3%,2% and 4 %. All of the errors are under the acceptable range.

5.2. Model validation for isothermal single cell cold start

The initial membrane water content should be low, and the applied load current should be prominent to realize a successful PEM fuel cell cold start. The survival time will increase with low membrane water content, while the heat production will increase at the large load current. Therefore, validating the cold start model accuracy at a relatively large current density is necessary. Then another three operation conditions are conducted. The working conditions are initial membrane water content 4, applied current density 200 mA/cm²; initial membrane water 4, applied current density 400 mA/cm²; initial membrane water 4, applied current density 500 mA/cm².

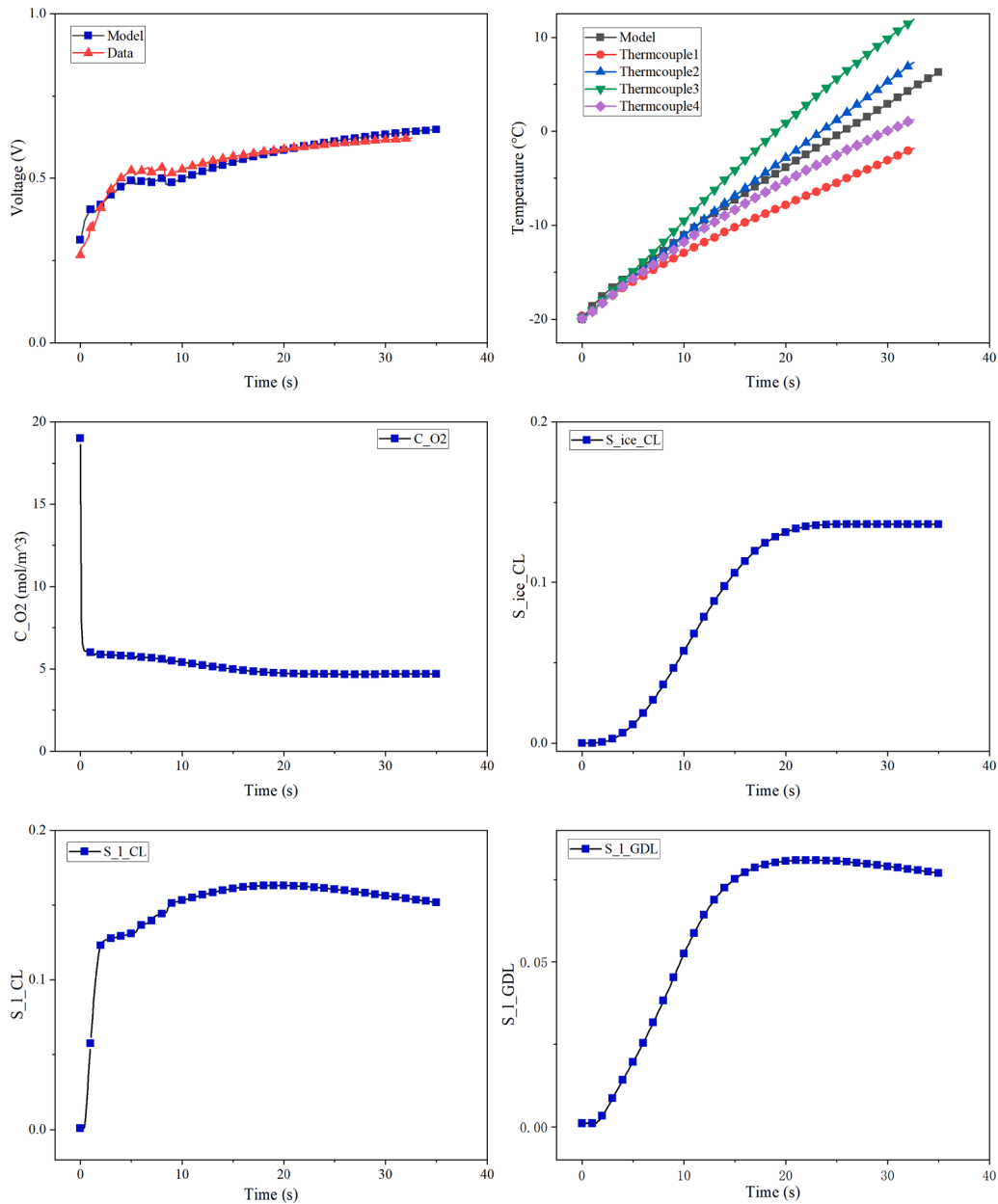


Fig. 9. Model dynamical result compared with experiment measurement and model prediction oxygen concentration, ice saturation and liquid water saturation.

The validation results of the cold-start model prediction voltage with experiment dates at large load current conditions are shown in Fig. 6. As we can see, the model prediction voltage can capture the experiment output voltage well. With increasing applied current density, the survival time of the cold start process decreases rapidly. When the load current density is 500 mA/cm^2 , only 7 s passed the cold start failed. From the voltage curves, it is clear that two stages exist in the cold start process. One is membrane absorption water, while another ice exists in CL and blocks oxygen transport. Compare the model and experiment voltage; the three condition errors are separately 2.5 %, 1.7 % and 3.3 % under the acceptable error range.

5.3. Result for stack non-isothermal cold start

In order to verify the model prediction accuracy for the stack-level cold start in a no-isothermal process, three operation conditions are considered here to validate the cold-start model. All of the three conditions considered here are separately constant current 400 mA/

cm^2 , 500 mA/cm^2 and current ramping $15 \text{ mA/cm}^2/\text{s}$. Model prediction voltage and temperature compared with experimental data are shown in Fig. 7 and Fig. 8.

From Fig. 7, we can see that the model prediction voltage can capture the experiment data. However, the error is relatively large for the constant current 400 mA/cm^2 and 500 mA/cm^2 . The active stack area is 330 cm^2 , and the current density distribution nonuniformity in the active area is much more severe than in a single cell. At the cold start initial stage, the model prediction voltage can capture the experimental data well, and then the model error is tiny. While in the failure stage, the ice distribution in the active area is different, and the ice-induced overpotential is also different. At the same time, the fuel cell active area works as a parallel form, and the worst performance part dominates the fuel cell output voltage. From Fig. 7, we can see that the model prediction cold start survival time error is almost 5 s and 3 s for the constant current condition. The current ramping condition model and experiment survival time are almost the same. The error of the model prediction voltage is under the acceptable range.

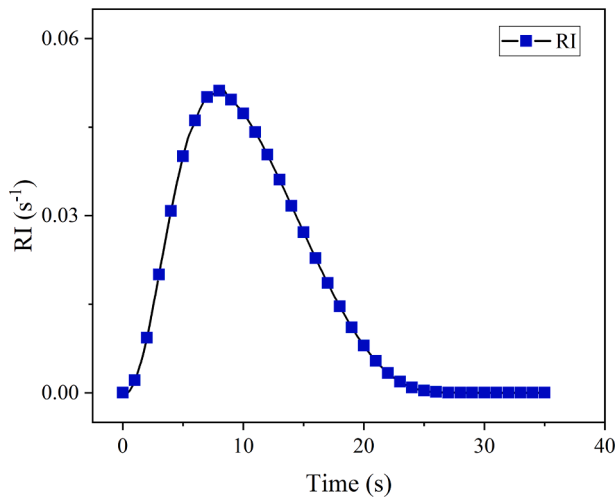


Fig. 10. Model prediction dynamical cold start process crystallization rate.

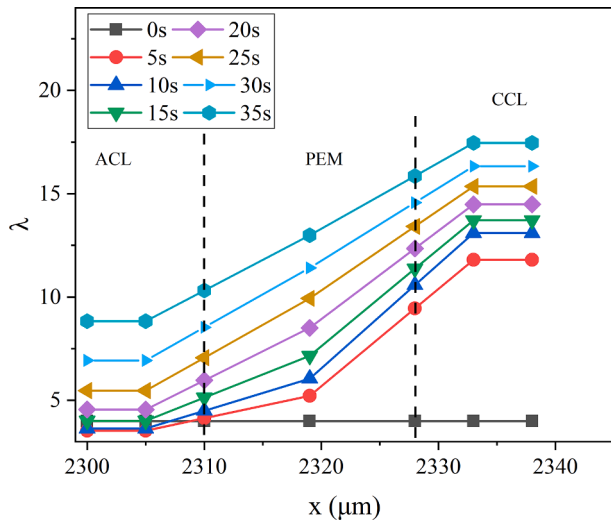


Fig. 11. Model prediction dynamical cold start process membrane water distribution.

Table 5

Model error analysis.

Fuel cell	Condition (mA/cm ² ; lambda)	Mean Voltage Relative Error	Mean Temperature Error (°C)
Single Cell	100,4	1.5 %	—
	200,4	5.36 %	—
	300,4	4.66 %	—
	400,4	1.71 %	—
	500,4	3.28 %	—
	100,5.7	0.8 %	—
	100,7.3	3.53 %	—
Stack	400,4	3.53 %	0.28
	500,4	2.23 %	1.12
	Ramp 15 mA/cm ² /s	5.86 %	1.05
	Dynamical step	0.43 %	0.69

While the comparison of model prediction temperature at the coolant between cell#1 and cell#2 with experiment measurement data are shown in Fig. 8. The model prediction temperature can capture the experiment measurement data well at the initial stage. While at the failure stage, the model temperature is inconsistent with experimental data, mainly due to the uneven current density distribution in the active

area. Meanwhile, heat power is dominated by voltage loss and current, and heat amount production and heat loss due to heat convection of different regions are not the same. So, the four thermocouples present different temperatures in the active area positions. However, the model prediction temperature is almost consistent with the average temperature of four thermocouples. Temperature errors of model and temperature are under the acceptable range.

5.4. Model validation for stack cold start

In Fig. 9, dynamical current density condition was conducted to study the stack cold start performance. The load current was 231A from 0 to 1.4 s, 247.5A from 1.4 to 5.4 s, 264A from 5.4 to 6.9 s, 280.5A from 6.9 to 8.4 s, and 305A from 8.4 to end. From Fig. 9, the model voltage can well reveal the dynamical load variation process. Meanwhile, the experiment measurement voltage can be captured accurately. From the model prediction temperature and thermocouples measurement data, the model can capture the cold start temperature variation trend. As we analyzed above, the large active area stack cold start, the current density distribution and the heat convection coefficient is different. The one-dimensional model can only reflect the average temperature well. Meanwhile, the thermocouples are displayed in the coolant channel, while the heat source mainly exists in MEA. So, the initial temperature measurement and model prediction is inconsistent.

During the dynamical cold start process, the oxygen concentration in CL, ice saturation in CL and liquid water saturation distribution in CL and GDL are displayed in Fig. 9. The ice volume fraction in GDL is small and not displayed here. According to the classical nucleation theory, the liquid water is first nucleated when the nucleation diameter is excess critical size, and the liquid water starts to crystallize to ice. The saturation membrane desorbed from ionomer during the cold start process and formed supercooled water in CL. Then the liquid water saturation rapidly increases in the CL while the liquid water is transported to GDL due to the capillary pressure. As Fig. 10 shows, the crystallization rate increases rapidly with the liquid water volume. However, as fuel cell temperature rises, the crystallization rate R.I. begins to decline. At 10 s, the temperature increased almost to -5°C , and the R.I. decreased almost to zero. Therefore, it is necessary to rapidly increase the fuel cell temperature to -10°C above to realize a successful cold start.

During the cold start process, the membrane water distribution in the MEA is displayed in Fig. 11. The initial membrane water content is set at four after equilibrium purge. The CCL ionomer membrane water content increases rapidly due to the electro-osmotic drag. Meanwhile, the ACL ionomer water content decreased. As the amount of water electro-osmotic drag is more significant than back-diffusion, the membrane water content in the cathode catalyst increases while the anode decreases. As time goes on, the back-diffusion water amount increases with the water content gradient. Then the anode water content in the catalyst layer increase. When CCL membrane water content reaches saturation, the membrane water content in CCL stays constant. As the fuel cell temperature rises, the maximum membrane water content increases. As Fig. 11 shows, CCL membrane water content reaches almost 16 at 30 s.

Finally, the cold start model voltage and temperature errors of all operating conditions are listed in Table 5. All errors are within the acceptable engineering range. The result demonstrates the prediction capabilities of the cold start model. The reduced and simplified model balances the low computation cost and fidelity of the numerical simulation model. The model can reproduce the behaviour that much more computationally intensive numerical models present. The efficiency and fidelity of the simplification model show a significant advantage for real-time estimation and task online optimization control.

6. Conclusion

This study presents a reduced and simplified cold-start model of PEM fuel cells. The sparse space discrete method achieves model fidelity and

high computational efficiency, and the computational time is at least an order of magnitude faster than the numerical model. Notably, a simple ice-crystallization kinetic expression is incorporated into the cold start model to simulate the ice formation in CL and the cold start failure process. An analytical method is used to capture the utilization efficiency for current density distribution unevenly in CL, leading to the effective ohmic resistance that is different from the whole ionomer resistance. Then a parameters identification approach is developed according to the single-cell cold start experiment and stack cold start experiment data. Furthermore, the model is validated against various operating conditions for isothermal and non-isothermal cold start conditions. Then non-isothermal dynamical cold start experiment results also are verified to show the model's capability to capture the fuel cell voltage and temperature variation. The validation results show that the simplified model has high fidelity and computational efficiency and is suitable for real-time estimation and control for vehicle PEMFC cold start. Based on this simplified model, real-time estimation of the internal state of the low-temperature cold start process of the onboard fuel cell system and optimal control of the cold start strategy will be the next primary research direction. However, this simplified model requires parameter calibration for different fuel cell systems to verify the model's accuracy, which has certain limitations in the application process.

CRediT authorship contribution statement

Jianjian Tao: Methodology, Software, Conceptualization, Writing – original draft. **Xuezhe Wei:** Conceptualization, Methodology. **Pingwen Ming:** Visualization, Investigation. **Xueyuan Wang:** Writing – review & editing. **Shangfeng Jiang:** Funding acquisition, Data curation. **Haifeng Dai:** Writing – review & editing, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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